



8328 - Direct detection of a multi-planet system caught in formation

Cycle: 4, Proposal Category: GO

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JWST Proposal 8328 (Created: Friday, June 27, 2025, 11:00:22AM Eastern Standard Time) - Overview

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
Observation Folder				
	1	Science observation	NIRCam Coronagraphic Imaging	(1) UCAC4-300-090317
	2	PSF Reference Observation	NIRCam Coronagraphic Imaging	(2) HD-150178

ABSTRACT

With thousands of extrasolar planets detected in the last decades, we find ourselves asking the question how to explain the huge diversity in planet properties and system architectures. To understand the key processes involved in planet formation we must turn to the initial conditions in the dust and gas rich disks around young stars. Near-infrared and sub-mm observations have revealed a wealth of sub-structures in disks; smoking guns pointing to the presence of embedded planets. Yet the detection of these nascent planets has proven difficult, and until now, only a single system has been identified where multi-wavelength observations could unambiguously confirm the presence of planets. One major difficulty for the direct

imaging of these objects is that many of them are too faint to be detected even with the most advanced ground based facilities. The superior sensitivity of space-based observations with JWST now gives us the opportunity for a new observational window into the earliest phases of planet formation.

In this program we propose to directly detect the planets in an exceptional disk around a young low-mass star. The disk shows a rare combination of spiral and ring structures in near-infrared scattered light, which strongly indicates the presence of an embedded planet in the inner disk region (30 au). Furthermore complementary ALMA high resolution observations of the disk's gas point to a second planet in the outer disk (116 au). We show that based on existing data and dedicated simulations, both planets can be detected simultaneously with JWST/NIRCam at 4.4 micron. This presents us with the unique opportunity to directly study a multi-planet system in formation.

OBSERVING DESCRIPTION

This is a proposal to directly detect two forming planets embedded in the disk around a young nearby star. The presence of the planets has been inferred from extensive ground-based observations with VLT/SPHERE as well as ALMA. Furthermore we demonstrate that the planets can not be detected from the ground, while they are well detectable with JWST from space.

We are using NIRCam coronagraphic imaging in the F444W filter to detect both planet simultaneously. We include reference star observations which will allow us to subtract the residual light of the bright primary star, critical to achieve the required high contrast for the planet detection. We have vetted our reference star using VLT/SPHERE to exclude a close visual binary.

With ~6h total of JWST time we have the unique opportunity to directly study a multi-planet system in formation. The JWST data will be analysed in concert with the already existing ground-based data. Furthermore a companion proposal for higher spectral and spatial resolution follow-up observations with ALMA was already accepted (priv. communication).

Proposal 8328 - Targets - Direct detection of a multi-planet system caught in formation

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	UCAC4-300-090317	RA: 16 12 6.6799 (243.0278329d) Dec: -30 10 27.08 (-30.17419d) Equinox: J2000	Proper Motion RA: -12.746 mas/yr Proper Motion Dec: -26.97199995509436 mas/yr Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> UCAC4 300-090317 https://simbad.cds.unistra.fr/simbad/sim-id?Ident=%402378386&Name=UCAC4%20300-090317&submit=submit K=9.3 Category=Star Description=[Circumstellar disks, Exoplanet Systems, M stars, T Tauri stars, Young stellar objects]				
Fixed Targets	(2)	HD-150178	RA: 16 40 19.9849 (250.0832704d) Dec: -25 16 14.04 (-25.27057d) Equinox: J2000	Proper Motion RA: -21.281 mas/yr Proper Motion Dec: -1.5810000604687957 mas/yr Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> K=4.2 Category=Star Description=[A stars]				

Proposal 8328 - Observation 1 - Direct detection of a multi-planet system caught in formation

Observation	Proposal 8328, Observation 1: Science observation										Fri Jun 27 16:00:22 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Coronagraphic Imaging										
Diagnostics	(Science observation (Obs 1)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets											
	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
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	UCAC4 300-090317 https://simbad.cds.unistra.fr/simbad/sim-id?Ident=%402378386&Name=UCAC4%20300-090317&submit=submit K=9.3 Category=Star Description=[Circumstellar disks, Exoplanet Systems, M stars, T Tauri stars, Young stellar objects]										
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	1 UCAC4-300-090317	F210M	BRIGHT (ND Square)	SHALLOW4	65	1	1	59.155	227661	
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern		
	A		MASK210R		true		SUB640A210R		9-POINT-CIRCLE		
Confirmation	#	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time	Conf. Total Dithers				
	1	RAPID	6	1	1	64.421	1				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F200W	F444W	SHALLOW4	6	10	9	90	11303.611	174414	

Proposal 8328 - Observation 1 - Direct detection of a multi-planet system caught in formation

PSF References	PSF Reference Observation (Obs 2) (PSF Reference; Filters [F200W/F444W]) Additional Justification: false
Special Requirements	No Parallel Attachments Group Observations 1, 2, Non-interruptible

Proposal 8328 - Observation 2 - Direct detection of a multi-planet system caught in formation

Observation	Proposal 8328, Observation 2: PSF Reference Observation										Fri Jun 27 16:00:22 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Coronagraphic Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	HD-150178	RA: 16 40 19.9849 (250.0832704d)			Proper Motion RA: -21.281 mas/yr					
			Dec: -25 16 14.04 (-25.27057d)			Proper Motion Dec: -1.5810000604687957 mas/yr					
			Equinox: J2000			Epoch of Position: 2000					
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	K=4.2										
	Category=Star										
	Description=[A stars]										
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	F210M	BRIGHT (ND Square)	RAPID	33	1	1	6.193	227661	
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern		
	A		MASK210R		false		SUB640A210R		9-POINT-CIRCLE		
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F200W	F444W	RAPID	6	2	9	18	527.784		
PSF References	PSF Reference: true										

Proposal 8328 - Observation 2 - Direct detection of a multi-planet system caught in formation

Special Requirements	Group Observations 1, 2, Non-interruptible
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